

321. Let O be the origin, the point A be $z_1 = \sqrt{3} + 2\sqrt{2}i$, the point B(z_2) be such that $\sqrt{3}|z_2| = |z_1|$ and $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$. Then :

(A) ABO is an obtuse angled isosceles triangle (B) ABO is a scalene triangle

(C) area of triangle ABO is $\frac{11}{4}$ (D) area of triangle ABO is $\frac{11}{\sqrt{3}}$

माना कि O मूल बिन्दु, बिन्दु A, $z_1 = \sqrt{3} + 2\sqrt{2}i$, बिन्दु B(z_2) इस प्रकार हैं कि $\sqrt{3}|z_2| = |z_1|$ और $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$ है। तब :

(A) ABO एक अधिककोण समद्विबाहु त्रिभुज है। (B) ABO एक विषमबाहु त्रिभुज है।

(C) त्रिभुज ABO का क्षेत्रफल $\frac{11}{4}$ है। (D) त्रिभुज ABO का क्षेत्रफल $\frac{11}{\sqrt{3}}$ है।

Ans. A

Sol. $z_1 = \sqrt{3} + 2\sqrt{2}i$ & $\frac{|z_2|}{|z_1|} = \frac{1}{\sqrt{3}}$

$$\text{given } \arg\left(\frac{z_2}{z_1}\right) = \frac{\pi}{6}$$

$$z_2 = \frac{|z_2|}{|z_1|} \cdot z_1 e^{i\left(\frac{\pi}{6}\right)}$$

$$z_2 = \frac{1}{\sqrt{3}} \cdot \frac{(\sqrt{3} + 2\sqrt{2}i)(\sqrt{3} + i)}{2}$$

$$z_2 = \frac{(3 - 2\sqrt{2}) + i(2\sqrt{6} - \sqrt{3})}{2\sqrt{3}}$$

Now,

$$z_1 - z_2 = \frac{(3 + 2\sqrt{2}) + i(2\sqrt{6} - \sqrt{3})}{2\sqrt{3}}$$

$$|z_1 - z_2| = |z_2| \Rightarrow \Delta ABO \text{ is isosceles with angles } \frac{\pi}{6}, \frac{\pi}{6} \text{ \& } \frac{2\pi}{3}$$

322. Let $|z_1 - 8 - 2i| \leq 1$ and $|z_2 - 2 + 6i| \leq 2, z_1, z_2 \in \mathbb{C}$. Then minimum value of $|z_1 - z_2|$ is :

माना कि $|z_1 - 8 - 2i| \leq 1$ तथा $|z_2 - 2 + 6i| \leq 2, z_1, z_2 \in \mathbb{C}$ है। तब $|z_1 - z_2|$ का निम्नतम मान है :

(A) 7

(B) 3

(C) 10

(D) 13

Ans. A

Sol. $|z_1 - (8 + 2i)| \leq 1$

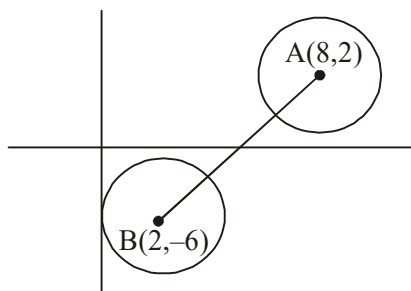
$C_1 : (8, 2)$

$r_1 = 1$

$|z_2 - (2 - 6i)| \leq 2$

$C_2 : (2, -6)$

$r_2 = 2$



$AB = 10$

$|z_1 - z_2|_{\min} = 10 - 2 - 1 = 7$

323. Let α and β be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0, z \in \mathbb{C}$. Then $4(\alpha^2 + \beta^2)$ is equal to :

माना समीकरण $(\bar{z})^2 + |z| = 0, z \in \mathbb{C}$ के सभी शून्येतर हलों के योग और गुणनफल क्रमशः α और β हैं। तो $4(\alpha^2 + \beta^2)$ बराबर है:

(A) 2

(B) 8

(C) 4

(D) 6

Ans. C

Sol. $z = x + iy$

$\bar{z} = x - iy$

$\bar{z}^2 = x^2 - y^2 - 2ixy$

$\Rightarrow x^2 - y^2 - 2ixy + \sqrt{x^2 + y^2} = 0$

$\Rightarrow x = 0 \quad \text{or} \quad y = 0$

$-y^2 + |y| = 0 \quad x^2 + |x| = 0$

$|y| = |y|^2 \quad x = 0$

$y = 0, \pm 1$

$\Rightarrow i, -i \quad \Rightarrow \alpha = i - i = 0$

are roots $\beta = i(-i) = 1$

$4(0 + 1) = 4$

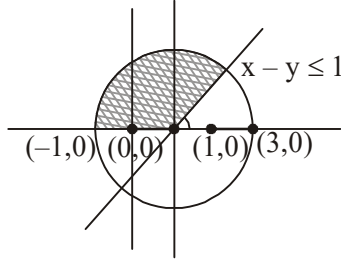
324. The area (in sq. units) of the region $S = \{z \in \mathbb{C} : |z - 1| \leq 2; (z + \bar{z}) + i(z - \bar{z}) \leq 2, \text{Im}(z) \geq 0\}$ is :

क्षेत्र $S = \{z \in \mathbb{C} : |z-1| \leq 2; (z + \bar{z}) + i(z - \bar{z}) \leq 2, \text{Im}(z) \geq 0\}$ का क्षेत्रफल (वर्ग इकाइयों में) है :

- (A) $\frac{7\pi}{4}$ (B) $\frac{7\pi}{3}$ (C) $\frac{17\pi}{8}$ (D) $\frac{3\pi}{2}$

Ans. D

Sol.



$$|z-1| \leq 2 \Rightarrow (x-1)^2 + y^2 = 4$$

$$(z + \bar{z}) + i(z - \bar{z}) \leq 2 \Rightarrow x - y \leq 1$$

$$\text{Im } |z| \geq 0 \Rightarrow y \geq 0$$

$$\text{Area of region} = \frac{1}{2}(2)^2 \left(\frac{3\pi}{4} \right) = \frac{3\pi}{2}$$

325. Consider the following two statements

Statement I : For any two non-zero complex numbers z_1, z_2 .

$$(|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq 2(|z_1| + |z_2|) \text{ and}$$

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that

$$\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}, \text{ then } \frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = 1.$$

Between the above two statements :

- (A) Statement I is correct but Statement II is incorrect
(B) Statement I is incorrect but Statement II is correct
(C) Both Statement I and Statement II are incorrect
(D) Both Statement I and Statement II are correct

कथनों

कथन I : किसी भी दो शून्येतर सम्मिश्र संख्याओं z_1, z_2 के लिए

$$(|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq 2(|z_1| + |z_2|) \text{ तथा}$$

कथन II : यदि x, y, z तीन भिन्न सम्मिश्र संख्याएँ हैं और a, b, c तीन धनात्मक वास्तविक संख्याएँ हैं ताकि $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$

है, तो $\frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = 1$ है।

में से :

- (A) कथन I सही है, परन्तु कथन II गलत है
 (B) कथन I गलत है परन्तु कथन II सही हैं
 (C) कथन I तथा कथन II दोनों गलत है
 (D) कथन I तथा कथन II दोनों सही है

Ans. A

Sol. (I) $(|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right|$

$$\left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq \left| \frac{z_1}{|z_1|} \right| + \left| \frac{z_2}{|z_2|} \right| = 1 + 1 = 2$$

(II) $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$

$$\frac{a^2}{(y-z)} \times \left(\frac{\overline{y-z}}{\overline{y-z}} \right) + \frac{b^2}{z-x} \times \frac{\overline{z-x}}{\overline{z-x}} + \frac{c^2}{x-y} \times \frac{\overline{x-y}}{\overline{x-y}}$$

$$= \frac{a^2}{|y-z|^2} (\bar{y} - \bar{z} + \bar{z} - \bar{x} + \bar{x} - \bar{y}) = 0$$

So not true

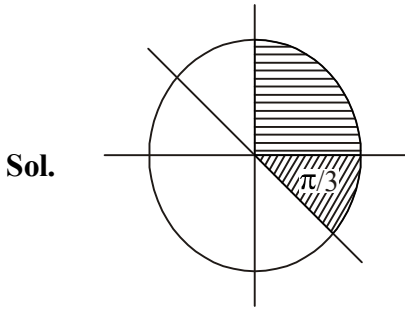
326. Let $S_1 = \{z \in \mathbb{C} : |z| \leq 5\}$, $S_2 = \left\{ z \in \mathbb{C} : \operatorname{Im} \left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i} \right) \geq 0 \right\}$ and $S_3 = \{z \in \mathbb{C} : \operatorname{Re}(z) \geq 0\}$. Then the area of the region $S_1 \cap S_2 \cap S_3$ is :

माना $S_1 = \{z \in \mathbb{C} : |z| \leq 5\}$, $S_2 = \left\{z \in \mathbb{C} : \operatorname{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \geq 0\right\}$ तथा $S_3 = \{z \in \mathbb{C} : \operatorname{Re}(z) \geq 0\}$ हैं। तो क्षेत्र $S_1 \cap S_2 \cap S_3$

का क्षेत्रफल है :

- (A) $\frac{125\pi}{24}$ (B) $\frac{125\pi}{4}$ (C) $\frac{125\pi}{12}$ (D) $\frac{125\pi}{6}$

Ans. C



$$S_1 : x^2 + y^2 \leq 5^2 ; S_2 : \sqrt{3}x + y \geq 0 ; S_3 : x \geq 0$$

$$\frac{z+1-i\sqrt{3}}{1-\sqrt{3}i} = \frac{((x+1)+i(y-\sqrt{3}))(1+i\sqrt{3})}{1+3}$$

$$\operatorname{Im}(z) = (x+1)\sqrt{3} + (y-\sqrt{3}) = \sqrt{3}x + y$$

$$S_1 \cap S_2 \cap S_3 = \frac{1}{4}\pi 5^2 + \frac{1}{2}5^2 \frac{\pi}{3} = \frac{125\pi}{12}$$

327. If z_1, z_2 are two distinct complex number such that $\left| \frac{z_1 - 2z_2}{\frac{1}{2} - z_1 \bar{z}_2} \right| = 2$, then :

- (A) both z_1 and z_2 lie on the same circle
 (B) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1
 (C) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$
 (D) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1

यदि z_1, z_2 दो भिन्न सम्मिश्र संख्याएँ इस प्रकार हैं कि $\left| \frac{z_1 - 2z_2}{\frac{1}{2} - z_1 \bar{z}_2} \right| = 2$ है, तो :

- (A) z_1 तथा z_2 दोनों समान वृत्त पर स्थित हैं
 (B) $z_1, \frac{1}{2}$ त्रिज्या के वृत्त पर तथा $z_2, 1$ त्रिज्या के वृत्त पर स्थित है
 (C) या तो $z_1, 1$ त्रिज्या के वृत्त पर या $z_2, \frac{1}{2}$ त्रिज्या के वृत्त पर स्थित है
 (D) या तो $z_1, \frac{1}{2}$ त्रिज्या के वृत्त पर या $z_2, 1$ त्रिज्या के वृत्त पर स्थित है

Ans. C

Sol. $\left| \frac{z_1 - 2z_2}{\frac{1}{2} - z_1 \bar{z}_2} \right| = 2$

$$\Rightarrow |z_1 - 2z_2| = |1 - 2z_1 \bar{z}_2|$$

$$\Rightarrow |z_1 - 2z_2|^2 = |1 - 2z_1 \bar{z}_2|^2$$

$$\Rightarrow |z_2|^2 - 2z_1 \bar{z}_2 - 2\bar{z}_1 z_2 + 4|z_2|^2 = 1 - 2\bar{z}_1 z_2 - 2z_1 \bar{z}_2 + 4|z_1|^2 |z_2|^2$$

$$\Rightarrow |z_1|^2 - 1 + 4|z_2|^2 - 4|z_1|^2 |z_2|^2 = 0$$

$$\Rightarrow (|z_1|^2 - 1)(1 - 4|z_2|^2) = 0$$

$$\text{either } |z_1|^2 - 1 = 0 \text{ or } 1 - 4|z_2|^2 = 0$$

$$|z_1| = 1 \quad \text{or} \quad |z_2| = \frac{1}{2}$$

$\therefore$ either z_1 lies on circle of radius 1

or z_2 lies on circle of radius $\frac{1}{2}$

328. Let z be a complex number such that $|z + 2| = 1$ and $\text{Im}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$. Then the value of $\left|\text{Re}\left(\overline{z+2}\right)\right|$ is :

माना z एक समिश्र संख्या इस प्रकार है कि $|z + 2| = 1$ और $\operatorname{Im}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$ तब $\left|\operatorname{Re}\left(\overline{z+2}\right)\right|$ का मान होगा :

- (A) $\frac{2\sqrt{6}}{5}$ (B) $\frac{24}{5}$ (C) $\frac{1+\sqrt{6}}{5}$ (D) $\frac{\sqrt{6}}{5}$

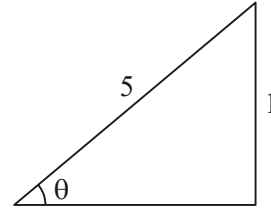
Ans. A

Sol. Let $z + 2 = e^{i\theta} \Rightarrow \left(\frac{z+1}{z+2}\right) = \frac{-2 + e^{i\theta} + 1}{e^{i\theta}} = 1 - \frac{1}{e^{i\theta}}$

$$\Rightarrow \left(\frac{z+1}{z+2}\right) = 1 - \cos \theta + i \sin \theta$$

$$\sin \theta = \frac{1}{5}$$

$$\operatorname{Re}(z+2) = \cos \theta \Rightarrow \left|\operatorname{Re}\left(\overline{z+2}\right)\right| = \frac{2\sqrt{6}}{5}$$



329. If the set $R = \{(a, b) : a + 5b = 42, a, b \in \mathbb{N}\}$ has m elements and $\sum_{n=1}^m (1 - i^{n!}) = x + iy$, where $i = \sqrt{-1}$, then the value of $m + x + y$ is :

यदि समुच्चय $R = \{(a, b) : a + 5b = 42, a, b \in \mathbb{N}\}$ में m अवयव है और $\sum_{n=1}^m (1 - i^{n!}) = x + iy$, जहाँ $i = \sqrt{-1}$, तब

$m + x + y$ का मान होगा :

- (A) 8 (B) 12 (C) 5 (D) 4

Ans. B

Sol. $a + 5b = 42, a, b \in \mathbb{N}$

$$a = 42 - 5b, b = 1, a = 37$$

$$b = 2, a = 32$$

$$b = 3, a = 27$$

$\vdots$

$$b = 8, a = 2$$

R has "8" elements $\Rightarrow m = 8$

$$\sum_{n=1}^8 (1 - i^{n!}) = x + iy$$

$$\text{for } n \geq 4, i^{n!} = 1$$

$$\Rightarrow (1 - i) + (1 - i^{2!}) + (1 - i^{3!})$$

$$= 1 - i + 2 + 1 + 1$$

$$= 5 - i = x + iy$$

$$m + x + y = 8 + 5 - 1 = 12$$

330. Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z - (6 + 8i)|$ is equal to :

माना z एक सम्मिश्र संख्या है, जिसके लिए $\frac{z-2i}{z+2i}$ का वास्तविक भाग शून्य है। तो $|z - (6 + 8i)|$ का अधिकतम मान बराबर है :

- (A) ∞ (B) 8 (C) 10 (D) 12

Ans. D

Sol. $\frac{z-2i}{z+2i} + \frac{\bar{z}+2i}{\bar{z}-2i} = 0$

$$z\bar{z} - 2i\bar{z} - 2iz + 4(-1) + z\bar{z} + 2zi + 2\bar{z}i + 4(-1) = 0$$

$$\Rightarrow 2|z|^2 = 8$$

$$\Rightarrow |z| = 2$$

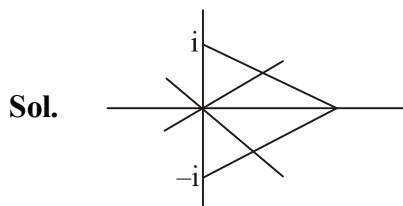
$$|z - (6 + 8i)|_{\text{maximum}} = 10 + 2 = 12$$

331. If $S = \{z \in \mathbb{C} : |z - i| = |z + i| = |z - 1|\}$, then $n(S)$ is :

यदि $S = \{z \in \mathbb{C} : |z - i| = |z + i| = |z - 1|\}$ है, तो $n(S)$ बराबर है :

- (A) 2 (B) 0 (C) 3 (D) 1

Ans. D



332. If $z = x + iy$, $xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to :

यदि $z = x + iy$, $xy \neq 0$ समीकरण $z^2 + i\bar{z} = 0$ को संतुष्ट करता है, तो $|z^2|$ बराबर है :

- (A) 1 (B) 1/4 (C) 9 (D) 4

Ans. A

Sol. $z = x + iy$, $x \neq 0$, $y \neq 0 \Rightarrow |z| \neq 0$

$$z^2 + i\bar{z} = 0$$

$$\Rightarrow z^2 = -i\bar{z}$$

$$\Rightarrow |z^2| = |-i\bar{z}| = |-i||\bar{z}|$$

$$\Rightarrow |z|^2 = |z| \Rightarrow |z|(|z| - 1) = 0$$

$$\Rightarrow |z| = 1$$

333. If z is a complex number, then the number of common roots of the equations $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to :

यदि z एक सम्मिश्र संख्या है, तो समीकरणों $z^{1985} + z^{100} + 1 = 0$ तथा $z^3 + 2z^2 + 2z + 1 = 0$ के उभयनिष्ठों मूलों की संख्या है :

- (A) 0 (B) 3 (C) 1 (D) 2

Ans. D

Sol. Given $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$

The roots of $z^{1985} + z^{100} + 1 = 0$ be ω and ω^2 and also satisfies $z^3 + 2z^2 + 2z + 1 = 0$

$\therefore \omega$ and ω^2 are common solutions

(where ω is cube root of unity)

$\therefore$ 2 solutions

334. Let z_1 and z_2 be two complex numbers such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$. Then $|z_1^4 + z_2^4|$ equals :

माना दो सम्मिश्र संख्याओं z_1 तथा z_2 के लिए $z_1 + z_2 = 5$ तथा $z_1^3 + z_2^3 = 20 + 15i$ हैं। तो $|z_1^4 + z_2^4|$ बराबर है :

- (A) $30\sqrt{3}$ (B) $25\sqrt{3}$ (C) $15\sqrt{15}$ (D) 75

Ans. D

Sol. $z_1 + z_2 = 5$

$$z_1^3 + z_2^3 = 20 + 15i$$

$$z_1^3 + z_2^3 = (z_1 + z_2)^3 - 3z_1z_2(z_1 + z_2)$$

$$z_1^3 + z_2^3 = 125 - 3z_1z_2(5)$$

$$\Rightarrow 20 + 15i = 125 - 15z_1z_2$$

$$\Rightarrow 3z_1z_2 = 25 - 4 - 3i$$

$$\Rightarrow 3z_1z_2 = 21 - 3i$$

$$\Rightarrow z_1 \cdot z_2 = 7 - i$$

$$\Rightarrow (z_1 + z_2)^2 = 25$$

$$\Rightarrow z_1^2 + z_2^2 = 25 - 2(7 - i)$$

$$\Rightarrow 11 + 2i$$

$$(z_1^2 + z_2^2)^2 = 121 - 4 + 44i$$

$$\Rightarrow z_1^4 + z_2^4 + 2(7 - i)^2 = 117 + 44i$$

$$\Rightarrow z_1^4 + z_2^4 = 117 + 44i - 2(49 - 1 - 14i)$$

$$\Rightarrow |z_1^4 + z_2^4| = 75$$

- 335.** Let $S = \{z \in \mathbb{C} : |z - 1| = 1 \text{ and } (\sqrt{2} - 1)(z + \bar{z}) - i(z - \bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that $|z_1| = \max_{z \in S} |z|$ and $|z_2| = \min_{z \in S} |z|$. Then $|\sqrt{2}z_1 - z_2|^2$ equals :

माना $S = \{z \in \mathbb{C} : |z - 1| = 1 \text{ तथा } (\sqrt{2} - 1)(z + \bar{z}) - i(z - \bar{z}) = 2\sqrt{2}\}$ है। माना $z_1, z_2 \in S$ के लिए $|z_1| = \max_{z \in S} |z|$ तथा $|z_2| = \min_{z \in S} |z|$ हैं। तो $|\sqrt{2}z_1 - z_2|^2$ बराबर है :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $(\sqrt{2} - 1)(z + \bar{z}) - i(z - \bar{z}) = 2\sqrt{2}$

$$(\sqrt{2} - 1)(2x) - i(2iy) = 2\sqrt{2}$$

$$(\sqrt{2} - 1)x + y = \sqrt{2}$$

and circle $|z - 1| = 1$

$$(x - 1)^2 + y^2 = 1$$

$$(x - 1)^2 + \left(\sqrt{2} - (\sqrt{2} - 1)x\right)^2 = 1$$

$$(x - 1)^2 + \left(\sqrt{2}(1 - x) + x\right)^2 = 1$$

$$(x - 1)^2 + 2(1 - x)^2 + x^2 + 2x\sqrt{2}(1 - x) = 1$$

$$3(x - 1)^2 + x^2 + 2\sqrt{2}x(1 - x) = 1$$

$$(-2\sqrt{2} + 4)x^2 + x(-6 + 2\sqrt{2}) + 2 = 0$$

$$x = 1 \text{ or } x = \frac{1}{2 - \sqrt{2}}$$

$$\text{for } x = 1 \Rightarrow y = 1$$

$$\text{and } x = \frac{1}{2 - \sqrt{2}} \Rightarrow y = \sqrt{2} - \frac{1}{\sqrt{2}}$$

$$\text{So, } |\sqrt{2}z_1 - z_2|^2 = \left| \frac{1}{\sqrt{2} - 1} + i - 1 - i \right|^2 = 2$$

336. If z is a complex number such that $|z| \geq 1$, then the minimum value of $\left| z + \frac{1}{2}(3 + 4i) \right|$ is :

यदि एक सम्मिश्र संख्या z के लिए $|z| \geq 1$ है, तो $\left| z + \frac{1}{2}(3 + 4i) \right|$ का न्यूनतम मान है :

(A) 3

(B) $\frac{5}{2}$

(C) 2

(D) $\frac{3}{2}$

Ans. Bonus

Sol. $p(z)$ is a point lying outside circle $x^2 + y^2 = 1$

$$Q\left(\frac{-3}{2}, -2\right)$$

$$\left| z + \frac{1}{2}(3 + 4i) \right| = PQ$$

$(PQ)_{\min} = 0$ when P coincides with Q.

337. Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \{z \in \mathbb{C} : \operatorname{Re}(az^2 + bz) = a$ and $\operatorname{Re}(bz^2 + az) = b\}$ is equal to :

माना $a \neq b$ दो शून्येतर वास्तविक संख्याएँ हैं। तो समुच्चय $X = \{z \in \mathbb{C} : \operatorname{Re}(az^2 + bz) = a \text{ तथा } \operatorname{Re}(bz^2 + az) = b\}$ में अवयवों की संख्या है :

(A) 1

(B) 3

(C) 2

(D) 0

Ans. D

Sol. $\operatorname{Re}(az^2 + bz) = a$

$$az^2 + bz + a\bar{z}^2 + b\bar{z} = 2a$$

$$a(z^2 + \bar{z}^2) + b(z + \bar{z}) = 2a \quad \dots(1)$$

$$\operatorname{Re}(bz^2 + az) = b$$

$$bz^2 + az + b\bar{z}^2 + a\bar{z} = 2b$$

$$b(z^2 + \bar{z}^2) + a(z + \bar{z}) = 2b \quad \dots(2)$$

$$(1) \times b - (2) \times (a)$$

$$\Rightarrow (b^2 - a^2)(z + \bar{z}) = 0$$

$$\Rightarrow (z + \bar{z}) = 0 \quad (a^2 \neq b^2)$$

$$(1) \times a - (2) \times (b)$$

$$\Rightarrow (a^2 - b^2)(z + \bar{z}) = 2(a^2 - b^2) \quad (a^2 \neq b^2)$$

$$z^2 + \bar{z}^2 = 2$$

$$\Rightarrow (z + \bar{z})^2 - 2z\bar{z} = 2$$

$$z\bar{z} = -1$$

$$\Rightarrow 1 + 1^2 = -1$$

$$\Rightarrow \text{No solution}$$

$$\text{But when } a = -b,$$

$$\operatorname{Re}(az - az) = a$$

$$\Rightarrow \operatorname{Re}(a(x^2 - y^2 + i2xy) - a(x + iy)) = a$$

$$\Rightarrow a(x^2 - y^2) - ax = a$$

$$\Rightarrow x^2 - y^2 - x = 1$$

$$\Rightarrow x^2 - x - 1 = y^2$$

For any real values of y there two values of x , hence infinite complex numbers are possible.

338. Let the complex number $z = x + iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to :

माना सम्मिश्र संख्या $z = x + iy$ के लिए $\frac{2z-3i}{2z+i}$ मात्र काल्पनिक है। यदि $x + y^2 = 0$ है, तो $y^4 + y^2 - y$ बराबर है :

- (A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Ans. C

Sol. $z = x + iy$

$$\frac{(2z-3i)}{2z+i} = \text{purely imaginary}$$

$$\text{means } \operatorname{Re}\left(\frac{2z-3i}{2z+i}\right) = 0$$

$$\Rightarrow \frac{(2z-3i)}{(2z+i)} = \frac{2(x+iy)-3i}{2(x+iy)+i}$$

$$= \frac{2x+2yi-3i}{2x+i2y+i}$$

$$= \frac{2x+i(2y-3)}{2x+i(2y+1)} \times \frac{2x-i(2y+1)}{2x-i(2y+1)}$$

$$\operatorname{Re}\left[\frac{2z-3i}{2z+i}\right] = \frac{4x^2+(2y-3)(2y+1)}{4x^2+(2y+1)^2} = 0$$

$$\Rightarrow 4x^2+(2y-3)(2y+1) = 0$$

$$\Rightarrow 4x^2+[4y^2+2y-6y-3] = 0$$

$$\therefore x+y^2=0 \Rightarrow x=-y^2$$

$$\Rightarrow 4(-y^2)^2+4y^2-4y-3=0$$

$$\Rightarrow 4y^4+4y^2-4y-3=0$$

$$\Rightarrow y^4+y^2-y=\frac{3}{4}$$

Therefore, correct answer is option (3).

339. Let $S = \left\{ z = x + iy : \frac{2z-3i}{4z+2i} \text{ is a real number} \right\}$. Then which of the following is NOT correct :

माना $S = \left\{ z = x + iy : \frac{2z-3i}{4z+2i} \text{ एक वास्तविक संख्या} \right\}$ । तो निम्न में कौन सा सही नहीं है ?

(A) $y+x^2+y^2 \neq -\frac{1}{4}$

(B) $x = 0$

(C) $(x, y) = \left(0, -\frac{1}{2} \right)$

(D) $y \in \left(-\infty, -\frac{1}{2} \right) \cup \left(-\frac{1}{2}, \infty \right)$

Ans. C

Sol. $\therefore z = x + iy$ and $\frac{2z-3i}{4z+2i}$ is real number

$$\therefore \frac{2x + (2y - 3)i}{4x + (4y + 2)i} \text{ is real number.}$$

$$\therefore \frac{(2x + (2y - 3)i)(4x - (4y + 2)i)}{(4x)^2 + (4y + 2)^2} \text{ is real number}$$

$$\therefore -2x(4y + 2) + (2y - 3)4x = 0$$

$$\therefore x = 0$$

Here $x = 0$ and $y \in \mathbb{R}$ but $y \neq -\frac{1}{2}$ is not acceptable

in this case

Denominator will be zero.

- 340.** Let w_1 be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then the principal argument of $w_1 - w_2$ is equal to :

माना $z_1 = 5 + 4i$ को मूल बिंदु के सापेक्ष घड़ी की विपरीत दिशा में एक समकोण तक घुमाने पर बिंदु w_1 प्राप्त होता है तथा $z_2 = 3 + 5i$ को मूलबिंदु के सापेक्ष घड़ी की दिशा में एक समकोण तक घुमाने पर बिंदु w_2 प्राप्त होता है। तो $w_1 - w_2$ का मुख्य आयाम बराबर है :

(A) $-\pi + \tan^{-1} \frac{8}{9}$ (B) $\pi - \tan^{-1} \frac{33}{5}$ (C) $-\pi + \tan^{-1} \frac{33}{5}$ (D) $\pi - \tan^{-1} \frac{8}{9}$

Ans. D

Sol. $W_1 = z_1 i = (5 + 4i)i = -4 + 5i \dots (i)$

$$W_1 = z_2 (-i) = (3 + 5i)(-i) = 5 - 3i \dots (2)$$

$$W_1 - W_2 = -9 + 8i$$

$$\text{Principal argument} = \pi - \tan^{-1} \left(\frac{8}{9} \right)$$